Practical No 2: Jenkins Pipeline for File and Folder Operations

Objective

To perform file and folder operations using a Jenkins pipeline with multi-stage execution.

Description

This pipeline demonstrates multi-stage execution in Jenkins by creating a folder, creating a file inside it, writing content to the file, and displaying the content using post actions.

Prerequisites

1. Jenkins installed and running
2. Basic knowledge of pipeline syntax
3. Windows/Linux command understanding

Pipeline Script

```
pipeline {
agent any

stages {
stage('Project 1: Create Folder') {
steps {
echo "Project 1: Creating a folder 'myProjectFolder'"
// For Windows:
bat 'mkdir Folder1Jenkins'
}
}

stage('Project 2: Create File') {
steps {
echo "Project 2: Creating a file"
// For Windows:
bat 'type nul > Folder1Jenkins\myFile.txt'
}
}

stage('Project 3: Write Contents to File') {
```

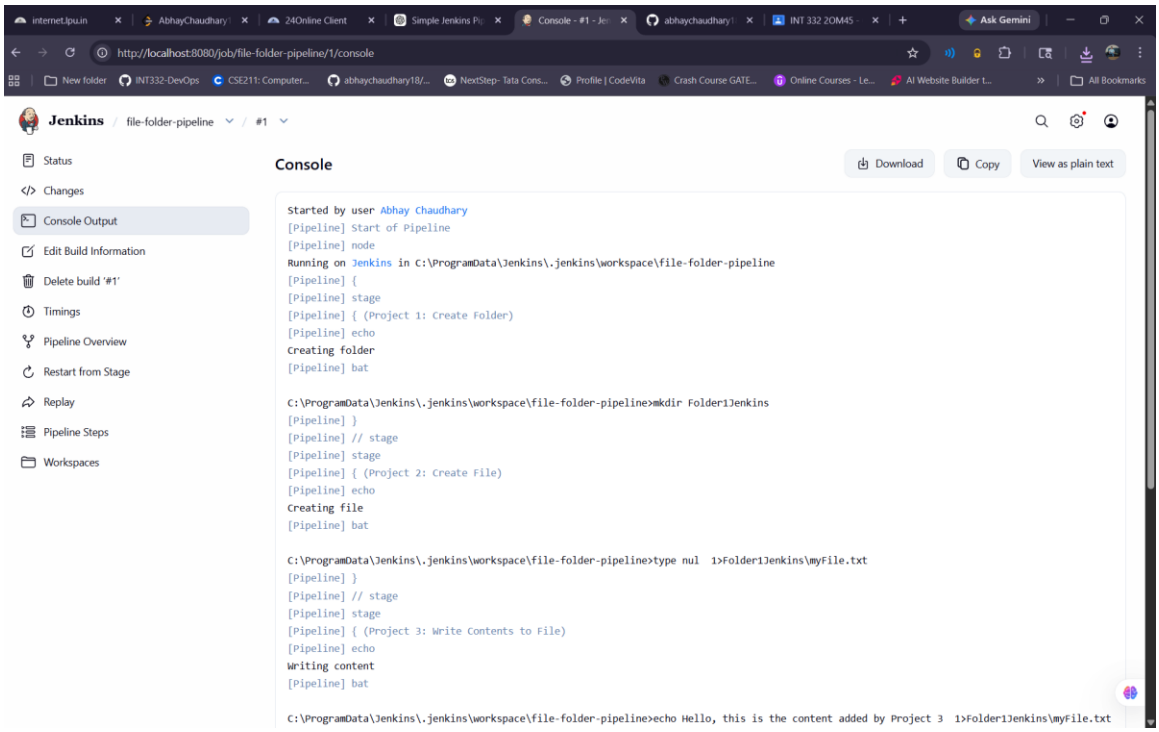
```
steps {
echo "Project 3: Writing contents into file"
// For Windows:
bat 'echo Hello, this is the content added by Project 3 > Folder1Jenkins\myFile.txt'
}
}
}

post {
always {
echo "Displaying the contents of the file:"
bat 'type Folder1Jenkins\myFile.txt'
}
}
}
```

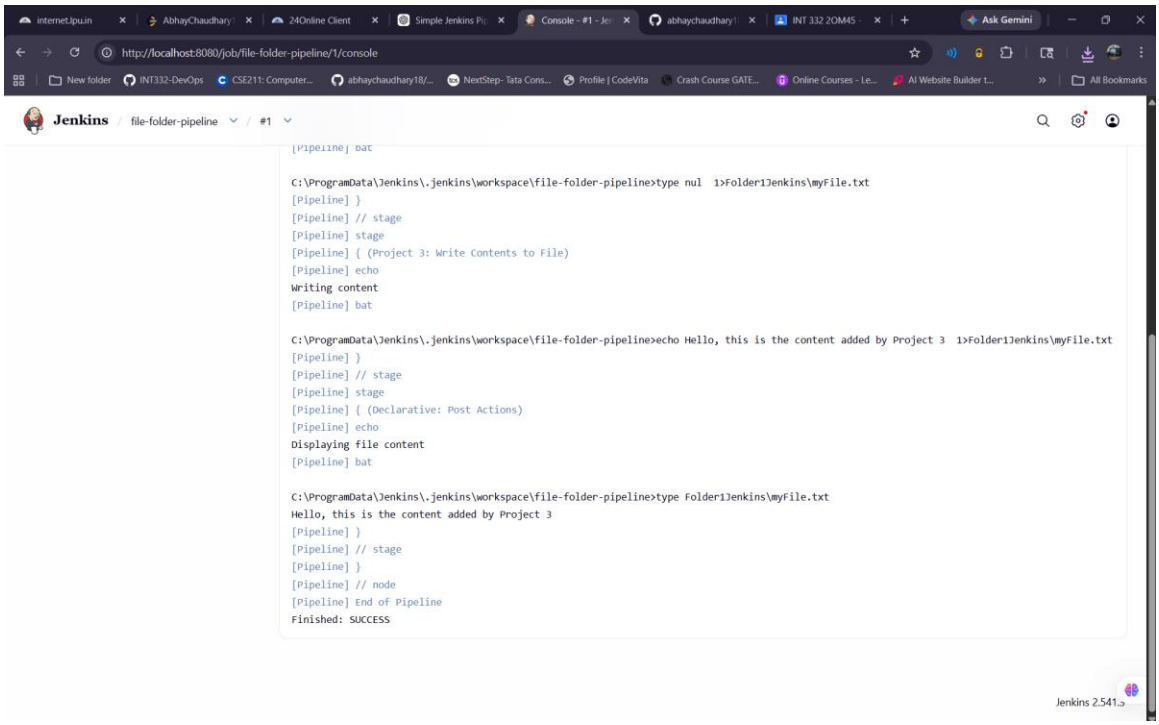
Expected Output

Project 1: Creating a folder
Project 2: Creating a file
Project 3: Writing contents into file
Displaying the contents of the file:
Hello, this is the content added by Project 3
Finished: SUCCESS

Conclusion

Successfully implemented a Jenkins pipeline with multiple stages to perform folder and file operations and verified output using post actions.


http://localhost:8080/job/file-folder-pipeline/1/console
Jenkins / file-folder-pipeline / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as plain text
Started by user Abhay Chaudhary
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\file-folder-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Project 1: Create Folder)
[Pipeline] echo
Creating folder
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\file-folder-pipeline>mkdir Folder1Jenkins
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Project 2: Create File)
[Pipeline] echo
Creating file
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\file-folder-pipeline>type nul 1>Folder1Jenkins\myFile.txt
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Project 3: Write Contents to File)
[Pipeline] echo
Writing content
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\file-folder-pipeline>echo Hello, this is the content added by Project 3 1>Folder1Jenkins\myFile.txt


http://localhost:8080/job/file-folder-pipeline/1/console
Jenkins / file-folder-pipeline / #1
C:\ProgramData\Jenkins\.jenkins\workspace\file-folder-pipeline>type nul 1>Folder1Jenkins\myFile.txt
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Project 3: Write Contents to File)
[Pipeline] echo
Writing content
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\file-folder-pipeline>echo Hello, this is the content added by Project 3 1>Folder1Jenkins\myFile.txt
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Displaying file content
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\file-folder-pipeline>type Folder1Jenkins\myFile.txt
Hello, this is the content added by Project 3
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.541.3